MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) Regular EXAMINATION October 2025

Third SEMESTER

Discipline Specific Elective (DSE) - MG3DSECF5200 - TCP/IP AND NETWORK SECURITY (SPECIALIZATION : NETWORK SECURITY)
(2024 ADMISSION ONWARDS)

Duration: 2 Hours Maximum Marks: 70

Part A

Short Answer Type Questions
Answer any 10 questions
Each question carries 2 marks

1. Identifies sending/receiving process, ensures correct delivery.
2. Combining multiple signals/data streams for transmission over a single channel.
3. Interconnection of devices for resource and data sharing.
4. TCP: Connection-oriented, reliable; UDP: Connectionless, faster.
5. No read up, No write down, Simple Security Property.
6. Packet filtering, Stateful inspection, Proxy, Next-gen firewall.
7. Hashing, File permissions.
8. Faster encryption/decryption, less computation.
9. Authn: Identity verification; Authz: Access rights validation.
10. Cryptographic proof ensuring authenticity and integrity.
11. AH, ESP, IKE.
12. ESP provides encryption; AH provides authentication only.

Part B

Short Essay Type Questions
Answer any 6 questions

Each question carries 5 marks

13. A unique string of characters that identifies each computer using the Internet Protocol to communicate over a network.Unique identifier; IPv4/IPv6; identifies host/network; used for routing.
14. SMTP (send), POP3 (retrieve).
15. Eavesdropping, traffic analysis; attacker monitors data.
16. Monitors traffic, enforces policy, logging, filtering, security zones.
17. Ticket Granting Ticket (TGT) issued, used to access services securely.
18. CA: Issues digital certificates; RA: Verifies user identity.
19. TLS stronger; supports key exchange, message integrity.
20. SSH for remote login; TLS for data transfer; both use encryption and authentication.

Part C

Essay Type Questions
Answer any 2 questions
Each question carries 10 marks

21. Chinese wall model with figure-
22. Standard for public key certificates; ensures identity and trust.
23. Combines encryption (PGP) and digital certificates (S/MIME) for confidentiality, integrity, and authentication.